DATABASE MANAGEMENT SYSTEM

ER DIAGRAM & RELATIONAL SCHEMA DESIGN

ONLINE SHOPPING SYSTEM

Modelling the online shopping application as a complete Entity-Relationship diagram and converting it into a normalised relational schema.

Submitted By	Palle Sai Rithesh
Roll Number	2511021240013
Course	BCA Game Development
Subject	Database Management System
Assignment	Assignment 2 — ER Diagram & Relational Schema

01 TASK 1

Entities, Attributes & Primary Keys

The core entities of the Online Shopping System, carried forward from the first DBMS assignment, with their attributes and primary keys identified.

◆ CUSTOMER

Attribute	Data Type	Key	Description
customer_id	INT	**PK**	Unique identifier for each customer
name	VARCHAR(100)	—	Full name of the customer
email	VARCHAR(120)	—	Contact email, used for login
phone	VARCHAR(15)	—	Contact phone number
address	VARCHAR(255)	—	Delivery address

◆ CATEGORY

Attribute	Data Type	Key	Description
category_id	INT	**PK**	Unique identifier for each category
category_name	VARCHAR(80)	—	Name of the category, e.g. Electronics
description	VARCHAR(255)	—	Short description of the category

◆ PRODUCT

Attribute	Data Type	Key	Description
product_id	INT	**PK**	Unique identifier for each product
name	VARCHAR(150)	—	Name of the product
price	DECIMAL(10,2)	—	Selling price of the product
stock_qty	INT	—	Quantity available in inventory

◆ ORDER

Attribute	Data Type	Key	Description
order_id	INT	**PK**	Unique identifier for each order
order_date	DATETIME	—	Date and time the order was placed
status	VARCHAR(30)	—	Placed / Shipped / Delivered / Cancelled
total_amount	DECIMAL(10,2)	—	Total value of the order

◆ PAYMENT

Attribute	Data Type	Key	Description
payment_id	INT	**PK**	Unique identifier for each payment
payment_method	VARCHAR(30)	—	UPI / Card / Net Banking / COD
amount	DECIMAL(10,2)	—	Amount paid
payment_status	VARCHAR(30)	—	Pending / Paid / Failed / Refunded

◆ SHIPMENT

Attribute	Data Type	Key	Description
shipment_id	INT	**PK**	Unique identifier for each shipment
tracking_no	VARCHAR(50)	—	Unique courier tracking number
carrier	VARCHAR(60)	—	Delivery partner / courier name
delivery_status	VARCHAR(30)	—	Packed / Shipped / Out for delivery / Delivered

◆ REVIEW

Attribute	Data Type	Key	Description
review_id	INT	**PK**	Unique identifier for each review
rating	INT	—	Rating given by the customer, 1 to 5
review_text	VARCHAR(500)	—	Written feedback from the customer

> **Note:** *linking columns such as customer_id in Order or order_id in Payment are deliberately not listed as entity attributes here. At the ER level they are expressed as relationships (Section 2) and only appear as foreign keys once the schema is derived in Section 4.*

02 TASK 2

Relationships & Cardinalities

Seven relationships connect the seven entities above. Six are one-to-many; one — Order and Product — is many-to-many and is resolved into its own table in Section 4.

Relationship	Participating Entities	Cardinality	Meaning
PLACES	Customer → Order	**1 : M**	A customer may place many orders; each order is placed by exactly one customer.
BELONGS TO	Category → Product	**1 : M**	A category can group many products; each product belongs to exactly one category.
CONTAINS	Order ↔ Product	**M : N**	An order can include many products, and a product can appear in many orders. Carries its own attributes (quantity, unit_price).
PAID BY	Order → Payment	**1 : M**	An order can have more than one payment record, e.g. retries or refunds; each payment belongs to one order.
SHIPPED AS	Order → Shipment	**1 : M**	An order can be fulfilled through one or more shipments; each shipment belongs to exactly one order.
WRITES	Customer → Review	**1 : M**	A customer can write many reviews; each review is written by exactly one customer.
RECEIVES	Product → Review	**1 : M**	A product can receive many reviews; each review refers to exactly one product.

> **Note:** *no one-to-one (1:1) relationship occurs naturally in this domain — every association here either links one record to many related records, or (in the case of CONTAINS) links many records on each side.*

03 TASK 3

Complete ER Diagram

Drawn in Chen notation — entities as rectangles, relationships as diamonds, attributes as ovals, and primary keys underlined.

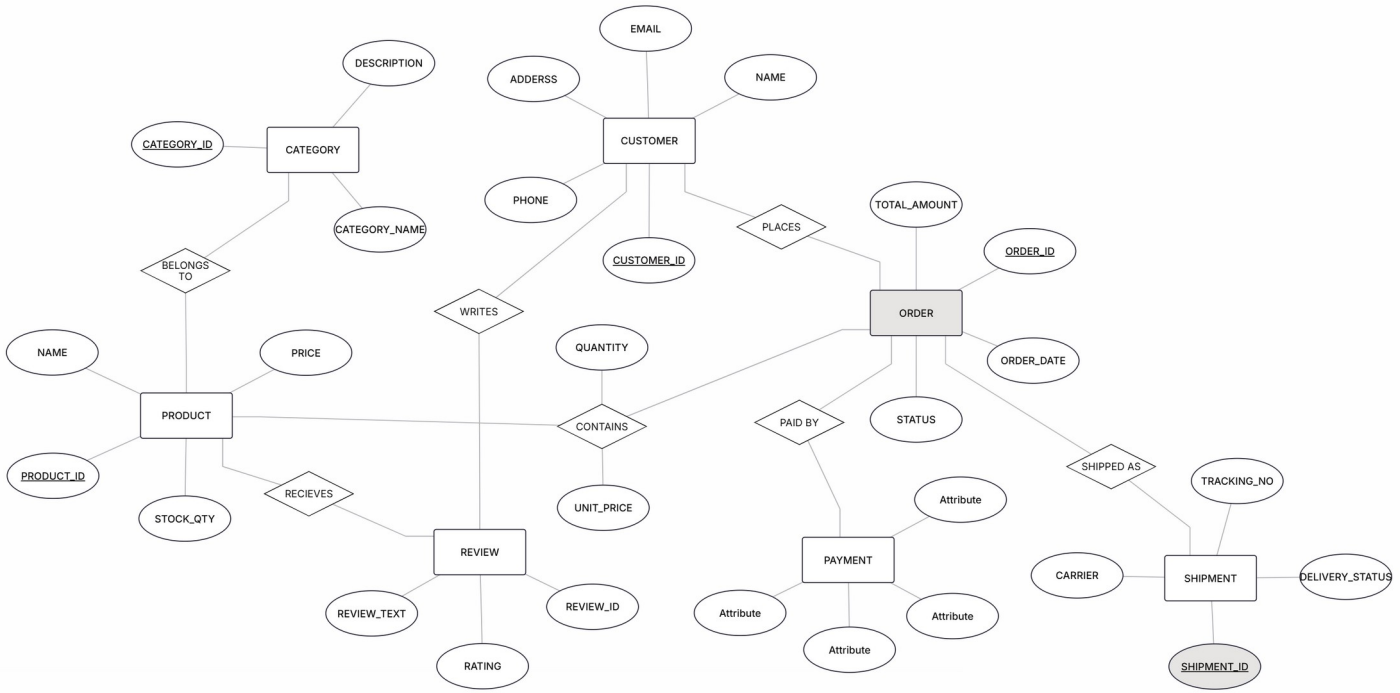


Figure 1 — Entity-Relationship Diagram of the Online Shopping System (Chen notation).

04 TASK 4

Relational Schema

The ER diagram is converted into tables using three standard mapping rules.

- **Entities become tables:** each entity turns into a table, and its attributes become columns; the entity's primary key stays the table's primary key.
- **1:M relationships become foreign keys:** the primary key of the "one" side is placed as a foreign key on the table of the "many" side (e.g. customer_id on Orders, order_id on Payments and Shipments).
- **The M:N relationship becomes a new table:** CONTAINS is converted into an associative table, Order_Items, holding the primary keys of both Orders and Products as foreign keys, plus the relationship's own attributes (quantity, unit_price) and its own surrogate primary key, order_item_id.

Figure 2: Relational Schema - Online Shopping System

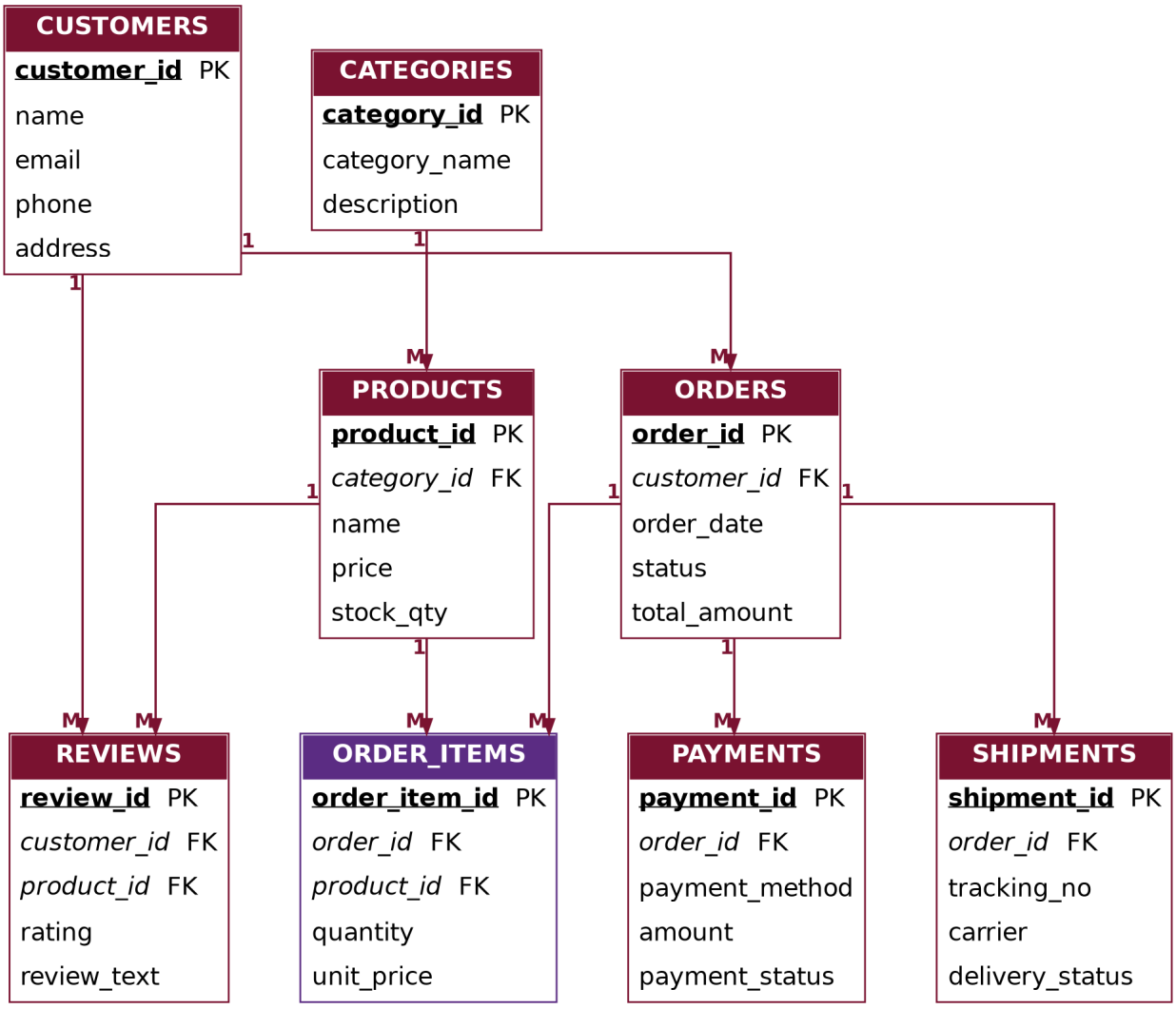


Figure 2 — Relational Schema of the Online Shopping System, showing primary and foreign keys.

◆ CUSTOMERS

Column	Type	Key	Description
customer_id	INT	**PK**	Unique customer identifier
name	VARCHAR(100)	—	Customer full name
email	VARCHAR(120)	—	Login / contact email (unique)
phone	VARCHAR(15)	—	Contact number
address	VARCHAR(255)	—	Delivery address

◆ CATEGORIES

Column	Type	Key	Description
category_id	INT	**PK**	Unique category identifier
category_name	VARCHAR(80)	—	Category name, e.g. Electronics
description	VARCHAR(255)	—	Short category description

◆ PRODUCTS

Column	Type	Key	Description
product_id	INT	**PK**	Unique product identifier
category_id	INT	**FK**	References Categories.category_id
name	VARCHAR(150)	—	Product name
price	DECIMAL(10,2)	—	Selling price
stock_qty	INT	—	Available inventory quantity

◆ ORDERS

Column	Type	Key	Description
order_id	INT	**PK**	Unique order identifier
customer_id	INT	**FK**	References Customers.customer_id
order_date	DATETIME	—	Date and time of order
status	VARCHAR(30)	—	Placed / Shipped / Delivered / Cancelled
total_amount	DECIMAL(10,2)	—	Total order value

◆ ORDER_ITEMS (RESOLVES THE M:N RELATIONSHIP)

Column	Type	Key	Description
order_item_id	INT	**PK**	Unique order-line identifier
order_id	INT	**FK**	References Orders.order_id
product_id	INT	**FK**	References Products.product_id
quantity	INT	—	Number of units of that product
unit_price	DECIMAL(10,2)	—	Price captured at the time of purchase

◆ PAYMENTS

Column	Type	Key	Description
payment_id	INT	**PK**	Unique payment identifier
order_id	INT	**FK**	References Orders.order_id
payment_method	VARCHAR(30)	—	UPI / Card / Net Banking / COD
amount	DECIMAL(10,2)	—	Amount paid
payment_status	VARCHAR(30)	—	Pending / Paid / Failed / Refunded

◆ SHIPMENTS

Column	Type	Key	Description
shipment_id	INT	**PK**	Unique shipment identifier
order_id	INT	**FK**	References Orders.order_id
tracking_no	VARCHAR(50)	—	Unique courier tracking number
carrier	VARCHAR(60)	—	Delivery partner
delivery_status	VARCHAR(30)	—	Packed / Shipped / Out for delivery / Delivered

◆ REVIEWS

Column	Type	Key	Description
review_id	INT	**PK**	Unique review identifier
customer_id	INT	**FK**	References Customers.customer_id
product_id	INT	**FK**	References Products.product_id
rating	INT	—	Rating, 1 to 5
review_text	VARCHAR(500)	—	Written feedback

05 TASK 5

Relationship Explanation

A plain-language walk-through of each relationship in the ER diagram and why it carries the cardinality it does.

- **Customer places Order (1:M) —** each order is tied to exactly one customer, but a customer's order history can grow without limit, so the "many" side sits on Order.
- **Category classifies Product (1:M) —** a category such as Electronics groups many products, while every product is filed under a single category to keep the catalogue unambiguous.
- **Order contains Product (M:N) —** an order normally holds several products, and any given product is naturally bought in many different orders over time, so neither side can be capped at one. The quantity and unit_price of each line are properties of the pairing itself, not of the order or the product alone, which is why they sit on the relationship rather than on either entity.
- **Order paid by Payment (1:M) —** one payment usually settles an order, but the model allows more than one payment row per order to capture retries, partial payments or refunds, without ever letting one payment settle two orders.
- **Order shipped as Shipment (1:M) —** most orders travel in a single shipment, but large or split orders may need more than one, so an order can have several shipment records while each shipment still belongs to one order.
- **Customer writes Review (1:M) —** a customer may review several purchases over time, but any one review is authored by a single customer.
- **Product receives Review (1:M) —** a popular product accumulates many reviews from different buyers, while each review naturally evaluates just one product.

06 SUMMARY

Conclusion

How the design fulfils Assignment 2's requirements.

The Online Shopping System is modelled as seven entities connected by seven relationships. Six relationships are one-to-many; the Order–Product relationship is many-to-many and, following standard ER-to-relational conversion rules, is resolved into its own associative table, Order_Items. This mirrors exactly how the first assignment's schema was built, confirming that the conceptual ER design and the working relational schema describe the same eight-table database.

Every table carries a single-column surrogate primary key, and every one-to-many relationship is implemented as a foreign key on the "many" side, keeping referential integrity straightforward to enforce and the design easy to extend.

Expected Output	Status
List of entities and attributes	✔️ **Included**
ER Diagram	✔️ **Included**
Relationships and cardinalities	✔️ **Included**
Relational Schema	✔️ **Included**
Primary and Foreign Key identification	✔️ **Included**
Brief relationship explanation	✔️ **Included**